TASK 03: CHOOSE VARIABLES AND NAME THE CONSTANTS

Course: CSA08 – Python Programming
Assignment: 1 – Payment Calculator for a Village Milk Co-operative
Bloom's Level: K3 – Apply
Marks: 3

1. Aim

To define meaningful variables and named constants for the milk payment calculator and understand Python naming conventions, dynamic typing and multiple assignment.

2. Naming Rules and Conventions

Python uses the following naming conventions:

- Variable names should be meaningful and written in lowercase.
- Multiple words in variable names are separated using underscores.
- Constants are generally written in uppercase letters.
- Variable names cannot begin with a number.
- Spaces and special characters are not allowed in variable names.

Examples:

litres = 10

fat_percentage = 3.5

farmer_id = "F101"

BASE_RATE = 30.00

3. Named Constants

Constants store values that should remain unchanged during program execution.

The following constants are used in the milk payment calculator:

Constant	Value	Purpose
BASE_RATE	30.00	Basic rate per litre
FAT_STEP_VALUE	0.40	Value for each 0.1% fat step
SNF_STEP_VALUE	0.30	Value for each 0.1% SNF step
BASE_FAT	3.5	Standard fat percentage
BASE_SNF	8.5	Standard SNF percentage
MIN_FAT	2.0	Minimum allowed fat

MAX_FAT 7.0 Maximum allowed fat

Using named constants makes the program easier to understand and maintain. If the milk rate changes, the value can be updated in one place.

4. Why Python Has No True Constants

Python does not have a special keyword to declare a variable as a constant.

Instead, programmers use uppercase names to indicate that a value should not be changed.

For example:

BASE_RATE = 30.00

This is a naming convention. Python still allows the value to be changed, but programmers are expected to avoid modifying it.

5. Dynamic Typing

Python is dynamically typed. The same variable can store different data types during execution.

Example:

```
value = 10
print(type(value))

value = "Milk"
print(type(value))
```

Output:

```
<class 'int'>
<class 'str'>
```

The variable first stores an integer and later stores a string. Python automatically identifies the type.

6. Multiple Assignment and Swapping

Python allows multiple variables to be assigned in one statement.

Example:

```
a, b = 10, 20
print(a, b)

a, b = b, a
print(a, b)
```

Output:

10 20

20 10

The values of a and b are exchanged without using a temporary variable.

7. Alternative Concept and Complexity Analysis

Method 1: Using Magic Numbers

Magic numbers are values directly written inside expressions instead of using named constants.

Example:

rate = 30 + (fat - 3.5) / 0.1 * 0.40

If the